

Productizing AI: an enterprise guide to private AI infrastructure

How to navigate the complexity of building
your private cloud for AI with open source



Contents

Contents	2
Executive summary	3
Introduction	4
What is AI Infrastructure?	6
When to use public vs. private cloud?	9
Private cloud for AI with open source	13
Canonical's partner ecosystem	16
Conclusion	18

Executive summary

Driven by the need for predictable costs, digital sovereignty, and regulatory compliance, leading enterprises are increasingly adopting private or hybrid cloud deployment models for their AI infrastructure. However, managing the complexity of private AI infrastructure can be a challenging proposition. It requires a tightly integrated ecosystem spanning specialized silicon, networking, container orchestration, data, MLOps software, and more.

Open source software provides a transparent and flexible foundation for this architecture – particularly given the critical role of Kubernetes in AI workloads and the prevalence of open source software across the AI landscape. As such, using a unified, end-to-end open source stack represents the most logical path to enterprise-grade AI infrastructure.

Building AI infrastructure on Ubuntu enables organizations to take advantage of Canonical's extensive partner ecosystem – which includes industry leaders such as NVIDIA, Intel, and AMD – to solve the notorious hardware enablement problem and freely choose the optimal silicon for their use case. This guide provides an overview of AI infrastructure, examines the respective benefits and applications of public and private clouds for AI workloads, and outlines how to build a private cloud for AI using Canonical's open source stack.

Introduction

Traditional IT infrastructure is suboptimal for AI workloads. From compute to networking to storage, the demands of most modern AI workloads outstrip the capabilities of a typical data center. It should come as no surprise then that the early AI rush has happened primarily in the public clouds. After all, why risk investing in expensive on premises infrastructure when you could simply access GPUs on demand via hyperscalers, especially when the technology and use cases were still untested.

But fast-forward to today and enterprises are no longer just experimenting with AI, they are productizing it. This is driving a shift in the way that organizations approach AI infrastructure – and the direction of that shift is, at least in part, back towards private data centers.

Gartner® predicts that by 2028, over 20% of enterprises will run either AI training or inference workloads in local data centers, up from 2% in early 2025.¹ That is not to say that AI workloads are leaving hyperscaler environments altogether; rather, hybrid deployment models that span both public clouds and on premises infrastructure are emerging as the most effective paradigm.

The drivers behind this trend are fourfold: cost, regulatory compliance, sovereignty, and the proliferation of AI factories.

Cost

When productizing AI at scale, public cloud economics can reach a point of diminishing returns. Whereas the upfront costs are low with no need to invest in physical infrastructure, long-term costs can be high and unpredictable, especially with large volumes of data. The costs of storing and moving significant amounts of data within the public cloud is far from trivial.

Regulatory compliance

The AI landscape is coming under increasing scrutiny from legislators globally, putting new pressure on enterprises to ensure the auditability, transparency, and security of their AI infrastructure. However, in a public cloud, the ability to perform forensic-grade audits or to verify the provenance of training data is limited by the transparency policies of the provider, and compliance may not be possible in a multi-tenancy environment. To satisfy regulations such as FedRAMP, HIPAA and the EU AI Act, the most practical option is to bring AI workloads into local data centers where enterprises can implement comprehensive auditing and zero-trust data governance.

Sovereignty

For various reasons – geopolitical tensions, evolving regulations, and new security challenges – some organizations need full control over their data, their infrastructure, and where it is located. This means that these organizations cannot rely exclusively on public cloud providers since doing so creates dependency on foreign companies and hands over control of the infrastructure to external parties. You can learn more about cloud sovereignty [in our guide](#).

¹ Gartner, Market Guide for Hybrid AI Infrastructure, By Dennis Smith, Chandra Mukhyala, 7 May 2025. Gartner is a trademark of Gartner, Inc. and/or its affiliates.

AI Factories

An AI factory is a purpose-built computing infrastructure designed to industrialize the artificial intelligence lifecycle, enabling organizations to train, deploy, and operate AI models at scale. For modern enterprises, the ability to continuously process raw data into trained models through an AI factory is a key source of competitive advantage. Consequently, AI factories are becoming increasingly prevalent across industries.

While the AI factory concept is deployment agnostic – it can exist in the public cloud or on premises – private AI factories are emerging as the predominant model, largely for the reasons outlined above. With AI workloads running 24/7, owning the hardware makes more sense than paying hourly GPU rates; and keeping the AI factory on premises gives the organization complete control for regulatory compliance and sovereignty, while bringing the compute closer to the data.

It's clear that hybrid AI infrastructure is on the rise, and for good reason, but the approach is not without its difficulties. Most notably, deploying and operating a private data center is inherently more complex than simply utilizing public clouds. In public clouds, the hardware, networking, and tooling is typically taken care of by the cloud provider; but with private infrastructure, these are all challenges that enterprises need to address themselves. This guide will explore the key considerations and best practices for tackling these problems, and explain how to build a private cloud for AI with open source.

The first step is understanding what goes into AI infrastructure.

What is AI Infrastructure?

AI infrastructure is a nebulous term. It is often used interchangeably to refer to a variety of hardware and software, ranging from data center servers to GPUs to MLOps solutions. None of these is incorrect; however, they fail to paint the complete picture. The best way to define AI infrastructure is as follows:

AI infrastructure encompasses all of the hardware and software necessary to develop, deploy, and manage artificial intelligence workloads at scale.

Effective AI infrastructure is an integrated stack made up of numerous interconnected layers. The key components include:

- Storage
- Silicon compute
- Networking
- Operating system
- Container orchestration
- Data solutions
- MLOps software

Let's take a look at each of these layers in more detail.

Storage

AI workloads rely on enormous amounts of data, which in turn makes high-performance storage a critical building block for AI infrastructure. Throughout the AI lifecycle, you will need to store a variety of different data types, including raw and pre-processed data, training datasets, the model itself, and inference results. The storage system will need to provide the flexibility to support all these data types and workloads, along with high throughput, reliability, and extreme scalability to keep up with the vast data volumes.

Silicon compute

Specialized silicon chips are at the heart of AI infrastructure. Also known as AI accelerators, these chips provide the accelerated compute resources necessary to speed up machine learning and drive AI training and inference. The main categories of AI accelerators are as follows:

- **CPUs** (central processing units) – General-purpose processors. Although CPUs lack parallel throughput for large-scale AI, modern high-performance CPUs can serve as accelerators for lightweight AI tasks.
- **GPUs** (graphics processing units) – Processors with thousands of cores optimized for parallel processing. Originally used for images and videos, they are also ideally suited for AI workloads and have become the backbone of AI processing.
- **NPU**s (neural processing units) – Specialized processors optimized for deep learning and machine learning tasks. NPUs are optimized to efficiently handle neural network operations, and are most typically used in edge computing.
- **DPUs** (data processing units) – Smart network interfaces that offload data movement, security, and storage tasks from the CPU/GPU to optimize the flow of information in distributed AI clusters.

- **ASICs** (application-specific integrated circuits) – Customized AI accelerators that have been designed to perform one specific task, with the goal of delivering higher performance than general-purpose silicon. An example of an ASIC is Google’s **TPU** (Tensor Processing Unit), designed for neural network machine learning.

Networking

FAI training is a highly distributed process. Large models do not fit in a single server’s memory and instead tasks run across hundreds and thousands of nodes. To handle complex matrix operations in parallel, these nodes bypass kernel overhead using RDMA to share memory directly over the network.

This leads to unique challenges. For example, because the networking is synchronized by design, collective operations (like all-reduce, all-gather, and reduce-scatter) must finish across all GPUs before the next steps begins. This means that the entire cluster’s throughput is tethered to the slowest link, making “tail latency” (the 95th or 99th percentile of delays) a critical bottleneck; even a single delayed packet can stall thousands of GPUs, wasting resources and increasing job completion time (JCT).

Traditional IP over Ethernet networking is ill-equipped to address the demands of highly parallel applications like AI training. Specialized solutions include InfiniBand, which provides native support for RDMA strict performance requirements, and specially optimized Ethernet transport technologies, such as NVIDIA’s Spectrum-X, Ultra Ethernet or Scale-Up Ethernet, to address congestion and head-of-line blocking issues present with RDMA over converged Ethernet (RoCE).

Operating system

The operating system (OS) manages the underlying hardware and provides the platform for subsequent software layers. It is also responsible for the low level silicon drivers and firmware.

For AI infrastructure, the OS of choice is almost always Linux due to its stability, performance, and open source flexibility and ecosystem. The overwhelming majority of AI frameworks and tools were developed on Linux (and especially on Ubuntu), and Linux typically enjoys better driver support for AI accelerators. This leads to superior optimization across the stack.

Container orchestration

Complex AI projects rely on container orchestration for provisioning, operating, and managing modernized workloads at scale. This typically means Kubernetes. Indeed, Gartner® predicts that, by 2028, 95% of new AI deployments will use Kubernetes, up from less than 30% today.² By abstracting the underlying hardware, Kubernetes significantly reduces the complexity of managing AI systems. It enables organizations to easily allocate resources across AI workloads, scale on demand, and ensures portability across environments.

² Gartner, Magic Quadrant™ for Container Management, By Dennis Smith, Wataru Katsurashima, Tony Iams, Lucas Albuquerque, Michael Warrilow, Stephanie Bauman, 6 August 2025.
Gartner and Magic Quadrant are a trademark of Gartner, Inc. and/or its affiliates.

Data solutions

Every AI project needs data, and every AI infrastructure deployment needs a variety of data applications so that organizations can efficiently store and use data when building models. Data solutions include:

- **Database management systems (DBMS)** – Provide scalable, high performance storage and management for large volumes of structured, unstructured, and semi-structured data.
- **Data processing and data pipelines** – Handle extract, transform, and load (ETL) operations, preparing raw data for model consumption.
- **Data caching solutions** – Minimize latency for frequently accessed data, enhancing inference speed.
- **Vector databases** – Store information as vectors, enabling fast retrieval and similarity searches for unstructured, high-dimensional data which would be difficult to query with traditional methods.

MLOps software

Machine learning operations (MLOps) is like DevOps for machine learning. It is a set of practices that automates machine learning workflows, ensuring scalability, portability and reproducibility. These practices rely on software integrated into the AI infrastructure stack. The key functionalities and tools include:

- **Model training** – The core iterative process where the model learns patterns from the training data and is periodically evaluated against a validation set to tune performance and prevent overfitting.
- **Model versioning** – Tracks and manages different iterations of AI models, enabling you to reproduce and compare results, roll back versions, and ensure auditability.
- **Feature store** – A centralized repository of processed data, known as “features”, enabling organizations to consistently reuse the same data across AI projects.
- **Experiment tracking** – Logs experiments and tracks parameters to enable result comparisons and visualizations.
- **Orchestration and pipelines** – Develop and deploy scalable workflows, typically with containers on Kubernetes-based systems, using directed acyclic graphs (DAGs) to map out complex sequences of tasks.
- **Model serving** – Deploys trained models into a production environment and makes it accessible to external applications via an API.
- **Observability** – Tracks model performance in production and monitors for factors such as model accuracy decay and data drift (when the input data changes over time so that it no longer aligns with the training data).

When to use public vs. private cloud?

Now that you have a clear picture of the key AI infrastructure components, the next step is understanding when to use public versus private clouds. Both approaches have their advantages and disadvantages that make them better suited to certain use cases, and if you pursue a hybrid deployment, success hinges on choosing the right platform for the right workload.

Public clouds

The simplicity, scalability, and accessibility of public clouds make them the ideal entry point for most organizations exploring AI. For mature AI programs, public cloud infrastructure remains a critical component of a hybrid strategy, providing burst capacity and access to specialized hardware for high-intensity training runs.

Pros

- With no need to invest in expensive physical hardware, public clouds offer a low upfront cost and minimal barrier to entry.
- Leading public cloud providers give users easy access to a large number of AI tools, models, and managed services with none of the complexity of acquiring, implementing, or managing them in-house.
- It is easy to scale usage up or down on demand. You can add thousands of GPUs for massive training runs in a matter of minutes.

Cons

- Long term costs can be high – especially with high volume workloads – and potentially unpredictable with variable pricing.
- Application data grows exponentially as your AI application grows. Moving and storing massive datasets in the cloud requires careful architectural planning to avoid high egress costs and maintain long-term flexibility.
- Meeting regulatory compliance is challenging, especially at the age of data sovereignty. Ensuring that data is being stored, transmitted and processed in compliance with regulations might not be achievable in a multi-tenancy cloud environment.

Use cases

Rapid prototyping and experimentation

For startups and enterprise innovation labs, the public cloud is the ultimate sandbox for failing fast and iterating quickly without financial risk. Instead of waiting months for hardware procurement and installation, engineering teams can spin up powerful GPU instances in minutes to test a new hypothesis or try a dozen different open-source models. This pay-as-you-go utility model lowers the barrier to entry, allowing a small team to build a proof of concept (PoC) for a customer service chatbot or a visual recognition tool using the exact same infrastructure as a tech giant. If the experiment fails, they simply shut down the instance and stop paying; if it succeeds, they have a production-ready environment immediately available.

Handling spiky and unpredictable workloads

Public cloud infrastructure is uniquely suited for businesses with highly variable traffic patterns

where demand can explode overnight. Consider a retail e-commerce platform during “Black Friday” or a tax-filing service in April: these applications may need 100x their normal compute power for just a few days. Public clouds provide elasticity, automatically adding thousands of virtual CPUs or GPUs to handle the surge and then instantly removing them when traffic subsides. Attempting to build a private data center for this peak volume alone would be financially ruinous, as massive amounts of expensive hardware would sit idle and depreciate for the other 11 months of the year.

Access to state-of-the-art innovation

The public cloud offers cutting-edge hardware and proprietary models that are too expensive or complex for most organizations to build themselves. Only the largest hyperscalers (like Google, AWS, and Microsoft) can justify the spend to build supercomputers with tens of thousands of the latest NVIDIA H100s or TPU pods. By renting this infrastructure, a mid-sized logistics company can train a massive optimization model that would otherwise require a supercomputer. Furthermore, public clouds provide API access to proprietary frontier models and managed services. Companies can build powerful apps on top of these world-class brains immediately, taking advantage of billions of dollars of R&D without having to train a foundation model from scratch.

Private clouds

On premises AI infrastructure should typically be the go-to when productizing AI at scale: when the benefits of cost, performance, and control justify the high upfront investment and increased complexity.

Pros

- Private clouds ensure transparent, predictable OPEX; and an efficiently run, well-utilized private cloud is often more cost-effective for equivalent resources than public clouds.
- With private infrastructure, you do not share your resources with other tenants, and you will usually enjoy superior data and compute locality, leading to better performance.
- You have full control over your data, processing, storage, and access, so you can protect sensitive data and meet sovereignty requirements and minimal barrier to entry.

Cons

- Purchasing and provisioning new hardware carries high fixed costs, and the investment can be underutilized if the workloads don’t ramp up as expected.
- Deployment and operation requires dedicated team resources and technical skill.
- The ability to scale resources is capped by the hardware you own, and adding capacity requires procuring additional hardware.

Use cases

Highly regulated industries and data sovereignty

In industries like healthcare, finance, and the public sector, increasingly stringent compliance and data sovereignty requirements can make public cloud adoption legally or operationally impossible. Private AI infrastructure allows these organizations to establish a “sovereign data” environment where sensitive information, such as patient medical records, banking transaction logs, or national security intelligence, never leaves their physical control. For example, a hospital can run diagnostic AI on medical imaging locally, ensuring no patient data traverses the public internet, while government agencies can utilize air-gapped networks to analyze sensitive data without any risk of external leakage or vendor surveillance.

Industrial automation and edge computing

For manufacturing and autonomous systems, the primary drivers for private infrastructure are reliability and latency rather than just privacy. Factories and oil rigs often operate in environments with poor or non-existent internet connectivity, necessitating local AI servers that can process data instantly. A visual quality control system on an assembly line, for instance, must analyze high-bandwidth video to spot defects in milliseconds, while an autonomous vehicle must make life-or-death decisions instantly. In these scenarios, the delay (latency) introduced by sending data to a cloud server and waiting for a response is unacceptable; the compute power must reside physically at the "edge" where the data is.

Cost control and performance optimization

For organizations operating at a massive scale, the variable "rent" of public cloud GPUs can eventually exceed the fixed cost of owning the hardware. Private AI infrastructure appeals to companies that perform continuous, heavy-duty workloads, such as a retailer that must fine-tune its recommendation engine nightly or a research lab running complex simulations 24/7. By shifting from an operational expenditure (OpEx) model of paying per token or per hour to a capital expenditure (CapEx) model of owning the clusters, these organizations can achieve significant long-term savings. Furthermore, owning the infrastructure allows for granular control over hardware performance, enabling engineers to optimize power consumption and latency in ways that standardized public cloud instances do not permit.

Hybrid

Given the diverse applications of public and private AI infrastructure, it should come as no surprise that the strength of a hybrid strategy lies in its flexibility. By assigning the right workloads to the right platform, enterprises can benefit from the best of both worlds. Here are some examples:

Data tiering and selective privacy

A hybrid approach enables organizations to balance the advanced capabilities of public frontier models with the strict security of on-premises systems. In this tiered model, highly sensitive data – such as personally identifiable information (PII) or classified financial records – remains on local, private servers where it is processed by smaller, specialized models. Meanwhile, complex but non-sensitive queries (e.g. anonymized data) are routed to powerful public cloud models to take advantage of their superior reasoning and general knowledge. This setup ensures that the organization benefits from state-of-the-art intelligence without violating data sovereignty laws.

Resource elasticity and cloud bursting

Hybrid environments are the right solution for businesses with unpredictable traffic patterns, allowing them to solve the peak capacity problem without overspending. An organization can maintain a lean private infrastructure sized to handle its average daily workload, ensuring predictable costs for steady-state operations. However, during sudden spikes in demand, such as the holiday period for a retailer or tax season for a financial firm, the system automatically bursts the excess traffic to the public cloud. This setup prevents service crashes during critical windows while avoiding the massive capital expense of buying hardware that would sit idle for the rest of the year.

The training-inference lifecycle split

Many organizations use a hybrid strategy to optimize the different stages of the AI lifecycle, leveraging the unique strengths of both environments. For example, a company might rent massive, cutting-edge GPU clusters in the public cloud to train a foundation model from scratch; a process

that requires immense compute power but only for a few weeks or months. Once the model is trained and frozen, the weights are downloaded to the company's private, lower-cost infrastructure for deployment and daily inference. This approach treats the public cloud as a rental facility for heavy lifting and the private environment as a cost-efficient home for long-term operation.

Whether in-house, or with the help of a managed service provider, private AI infrastructure must be built, maintained, and operated. Therefore, it represents the lion's share of the complexity in any hybrid deployment. For that reason, the remainder of this guide will focus on best practices for building and running a private AI cloud with open source, using Canonical's AI stack as an example.

Private cloud for AI with open source

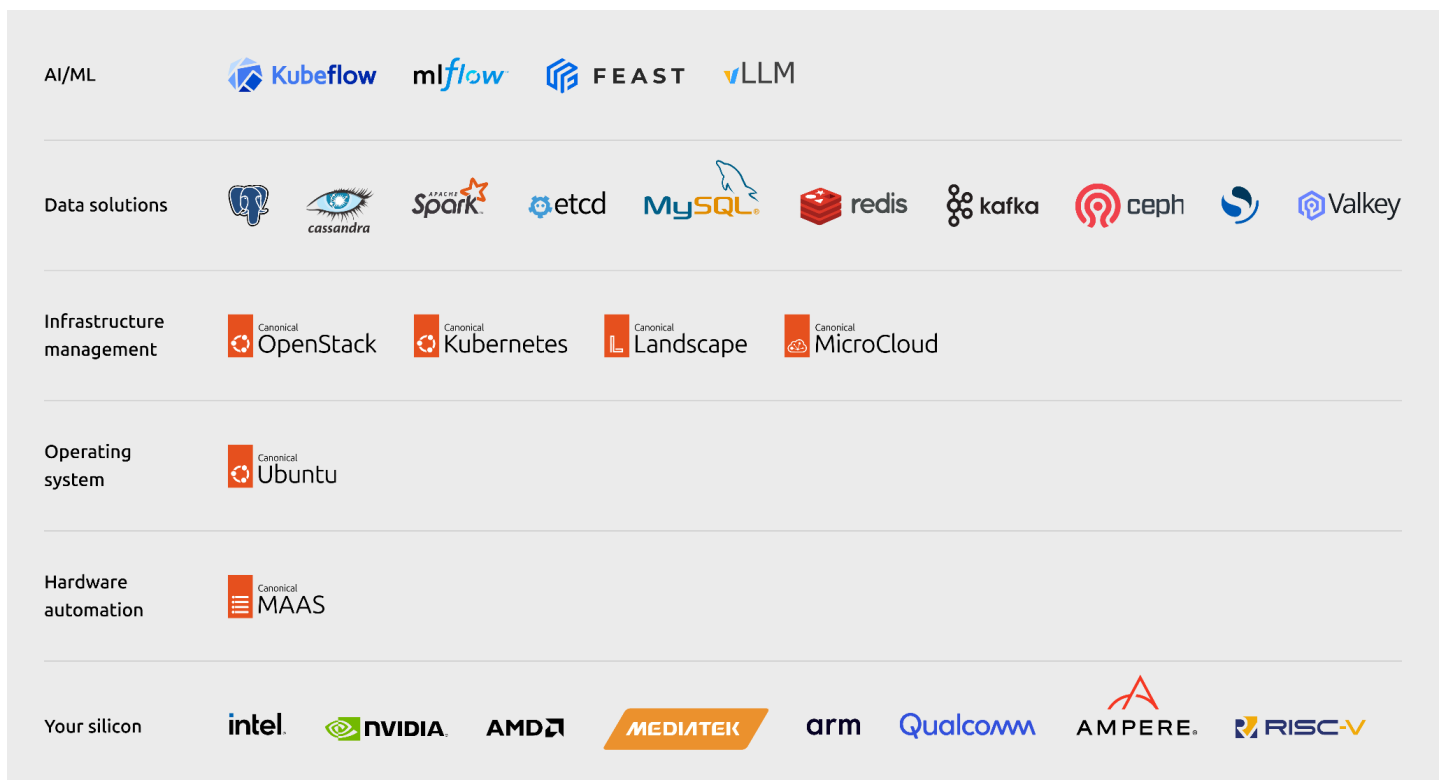
In this section, we will look at the benefits of building a private cloud for AI with open source software using Canonical's infrastructure portfolio.

Why use open source?

Why are enterprises choosing open source technology for their private AI infrastructure and AI factories?

Firstly, open source software offers the best path to sovereignty for AI infrastructure. You have full transparency and control over not just your data and models, but also the application source code. With everything hosted in your own private environment, you solve data residency concerns and make it significantly easier to manage access. In a similar vein, open source infrastructure lends itself to regulatory compliance. Satisfying regulations and auditing become significantly easier when you have full control and visibility over your software.

Open source solutions additionally help organizations remain hardware agnostic – enabling a greater degree of optimization and flexibility – and avoid potentially costly vendor lock-in.



Canonical's open source AI infrastructure stack.

We will cover all aspects of the private AI infrastructure stack, from hardware automation to the operating system, infrastructure management, databases and streaming solutions, and MLOps.

Data center automation

Minimizing operational complexity is essential to the long term viability of your AI infrastructure. This calls for automation throughout the stack, starting from the hardware provisioning and management of the compute nodes. MAAS is Canonical's solution for hardware automation. It enables organizations to provision physical servers as if they were cloud resources, and programmatically manage the entire hardware lifecycle as code using a cloud-like API. MAAS provides a central location to discover, commission, deploy, and repurpose machines, and makes large-scale infrastructure easier to operate thanks to repeatable workflows.

Ubuntu as the host OS

Even in a world of containers, it is essential not to overlook the host operating system. The host OS is the foundation that directly manages the hardware, and is responsible for ensuring that you can make the most of your high-value silicon resources. Ubuntu is engineered to run optimally across all major silicon architectures, so enterprises can mix-and-match best-of-breed hardware. What's more, Canonical collaborates with a network of partners to test and certify hardware with Ubuntu – more on this below.

Virtualize with OpenStack or MicroCloud

The cloud itself needs a virtualization platform, and OpenStack is by far the most popular open source option. OpenStack is challenging to operate, but [Canonical OpenStack](#), backed by Sunbeam, tames the complexity of the project to create an easy-to-consume, enterprise cloud solution.

For small-scale private clouds, edge environments, or resource-constrained use cases you can alternatively use [MicroCloud](#) – a low-touch, open source cloud solution for scalable clusters. MicroClouds are lightweight, can be deployed within minutes, and can scale up to 50 nodes.

Containerize with Kubernetes

As highlighted above in the component overview, containers and Kubernetes are fundamental elements of production-grade AI infrastructure. Kubernetes delivers the flexibility and modularity to support cloud-native AI workloads across the full AI lifecycle. At the same time, utilizing Kubernetes and containerization ensures portability of your workloads between environments in a hybrid cloud ecosystem.

Canonical's AI infrastructure stack is built around a performant, securely designed, opinionated, CNCF conformant distribution of Kubernetes. This distribution brings additional services and automation on top of the upstream, and up to 15 years of security maintenance and support.

The AI landscape is dominated by open source software, and Canonical maintains more than 2,000 widely used AI/ML libraries and tools, such as PyTorch, Tensorflow, Rapids, Triton, CASK, and many more. This includes a diverse portfolio of long-term supported containers for MLOps, data management and streaming applications – enabling you to build out the software layer of your AI infrastructure quickly and easily without worrying about open source maintenance challenges.

AI use cases and enterprise requirements are rarely one-size-fits-all, so you might find as you are scoping out your private cloud that you need to containerize open source packages that don't already exist in any off the shelf containers. Canonical can build and support custom containers for any open source package, tailored to your infrastructure and requirements. [Learn more about custom containers.](#)

Choose your data management stack

At a baseline, your AI infrastructure will need object storage (Ceph) and a key-value store (etcd). Depending on your use cases, you're likely to need Apache Kafka for data streaming, and potentially Apache Spark for data preprocessing. If you intend to use RAG, you'll require a vector database such as OpenSearch or PostgreSQL (with the pgvector extension). Canonical delivers all of these data solutions, and more, packaged as ready-to-use, fully OCI-compliant Docker containers – as well as debs and snaps for some applications.

End-to-end MLOps software

At the top of the stack, Canonical delivers all the open source applications you need to streamline the machine learning lifecycle, tightly integrated to ensure a smooth MLOps journey. The foundation of [Canonical MLOps](#) is Charmed Kubeflow, an enterprise-ready platform for orchestrating AI workflows. Charmed MLflow takes care of experiment tracking and managing the model lifecycle, and Charmed Feast is Canonical's enterprise-grade feature store.

The name "Charmed" refers to the fact that these solutions are deployed as charm bundles using Juju – an open source orchestration engine for software operators that enables the deployment, integration and lifecycle management of applications at any scale, on any infrastructure. The software operators are called charms: business logic encapsulated in reusable software packages that automate every aspect of an application's life. Whereas MAAS delivers automation at the hardware level, Juju and charms bring automation to the software layer.

Security, support, and cost-predictability

To handle production AI workloads, all this open source software needs enterprise-grade support for security and compliance. However, according to IDC research, 60% of organizations have only basic or no security controls to safeguard their AI/ML systems.³ With Canonical's solutions, the entire AI stack is supported under a single subscription, Ubuntu Pro – ensuring comprehensive coverage and predictable pricing. [Learn more about Ubuntu Pro.](#)

Next, let's take a closer look at the importance of a robust partner ecosystem.

³ <https://ubuntu.com/engage/2025-state-of-software-supply-chain>

Canonical's partner ecosystem

Productizing AI is challenging; moving from a prototype model to a reliable, scalable service requires careful engineering of complete systems. This shift makes the underlying AI hardware and operating system stability essential.

The Canonical silicon program directly addresses this by ensuring that the silicon chips integrate seamlessly within the software stack, from the OS all the way to data center orchestration systems, and edge deployments.

The partner ecosystem delivers benefits across three key areas:

1. De-risking heterogeneous AI compute

The core of AI compute is heterogeneity. No single vendor dominates all AI workloads (training, inference, edge). Enterprises face a "silicon chaos" of GPUs, NPUs, custom ASICs, and specialized accelerators. Canonical is a neutral arbiter, ensuring Ubuntu runs optimally across all major architectures (NVIDIA, AMD, Intel, Arm, Qualcomm, RISC-V). This enables enterprises to pick optimal hardware for their use cases without being locked into a single software ecosystem.

2. Accelerating time-to-value (go-to-production)

Time-to-market is often bottlenecked by the low-level work required to make a new chip work in a production environment. Ubuntu Certified Hardware is a trusted procurement signal for enterprises. Enterprises don't need to spend months validating if a new piece of hardware will work in their fleet. They know the hardware is validated for long-term support, and reliable AI operation. By providing standardized, pre-integrated secure boot enablement and firmware delivery, Canonical eliminates the need for OEMs, integrators, and, critically, the enterprise itself to perform custom OS engineering for every new piece of hardware. This accelerates the deployment of inference fleets, edge AI appliances, and private training clusters.

3. Enterprise-grade trust and long-term viability

As we explained earlier in this guide, an AI factory is a specialized environment designed to industrialize artificial intelligence and manage the entire end-to-end AI lifecycle. AI factories are long-term investments that demand the same stability, security, and lifecycle management as traditional mission-critical IT. Enterprises rely on the full LTS window guarantee provided by Canonical. This ensures that the low-level silicon drivers and firmware are maintained and receive timely security patches for the entire life of their AI infrastructure, protecting their investment from operating system-level vulnerabilities. Canonical owns the critical layer of security and reliability. This includes secure firmware delivery and continuous CI/CD validation, providing enterprise-grade guarantees that the AI platform is secure from the silicon up.

In essence, while enterprises focus on the exciting work of building AI systems that create market value, Canonical's partner ecosystem enables them to treat the complex, chaotic layer of hardware enablement as a solved problem. This converts silicon heterogeneity into a standardized, repeatable production platform.

In addition to these ongoing optimization, certification, and validation efforts, Canonical also collaborates with partners to develop reference architectures for key AI solutions, such as AI factories and confidential AI.

AI Factory reference architectures with partners

Canonical works with different silicon and hardware partners on reference architectures and validated designs. The NVIDIA Enterprise AI Factory, for instance, is a validated design for an AI factory that combines NVIDIA's own advanced hardware and software with solutions from its leading enterprise partners – including Canonical. The resulting design gives organizations a standardized, proven blueprint for building high-performing, scalable, and cost-effective on premises AI factories.

As part of the NVIDIA Enterprise AI Factory validated design, Canonical Kubernetes orchestration streamlines the process of building, deploying, and managing AI agents on high-performance infrastructure. It handles the burstable compute demands for training AI models and scales inference services for deployed agents based on real-time needs. Canonical Kubernetes additionally integrates with the NVIDIA GPUs and Network Operators to utilize NVIDIA hardware acceleration and supports the deployment of NVIDIA AI Enterprise, enabling AI workloads with NVIDIA NIM and accelerated libraries.

Organizations can also choose Ubuntu as the base OS for NVIDIA Enterprise AI Factory. Ubuntu is the most widely used OS for AI workloads, and using it as the base for AI factories gives enterprises a trusted repository for all their open source software.

Native support for confidential computing

Confidential AI uses confidential computing to protect the integrity of AI workloads at run-time. It can protect enterprises from insider threats, and vulnerabilities within a platform's privileged system software, while also opening up new business models.

Intel® Trust Domain Extensions (Intel TDX) is Intel's confidential computing technology. It creates hardware-isolated virtual machines – trusted domains – that protect data and applications from unauthorized access.

Taking full advantage of Intel TDX features requires an enabled software stack, which is why Canonical and Intel have an ongoing partnership to deliver Intel-optimized Ubuntu builds ahead of upstream, ensuring users can use the latest Intel TDX features as they are released. Ubuntu 25.10 shipped with native support for Intel TDX host capabilities, and this support will likewise be included out-of-the-box in Ubuntu 26.04 LTS, helping to drive mainstream adoption of confidential computing and confidential AI.

Conclusion

Relying exclusively on a single deployment model is rarely the optimal choice for modern AI workloads. While public clouds are essential for agility and experimentation, owning your own private AI infrastructure provides the predictability and performance optimization required for stable, high-volume workloads. And if you need sovereignty or regulatory compliance, there's no substitute for on-premises AI infrastructure. The mantra should always be: "right workload, right place."

Building your own private cloud for AI is not a trivial undertaking, but it does become more approachable when you use a single, end-to-end AI infrastructure stack. Focus on modernizing your workloads and building around Kubernetes, implement automation and orchestration, and choose a host OS that is optimized to work with your silicon.

Consulting services can also be a good way to accelerate your journey. Canonical's experts can deliver a workshop to upskill your infrastructure team, and work with you to define and implement a tailored private or hybrid cloud architecture.

Get in touch to learn more.

Further reading

[Report: Gartner® Predicts 2026: AI Sovereignty](#)

[Whitepaper: Sovereign cloud: the essential guide for enterprises](#)

[Whitepaper: AI-Powered Edge Networking with Dell, Intel, and Canonical](#)

[Guide: How Ubuntu enables on-premises confidential computing](#)